

Assignment #1

(Due on 14/4 at 11:59 PM)

1. Objective

The goal of this assignment is to evaluate how different image representation domains affect the performance of a standard Deep Neural Network.

You are required to train and evaluate a MobileNet-based classifier on images represented in multiple domains:

- Raw Spatial Domain (RGB images)
- Fourier Domain
- Wavelet Domain
- One additional domain of the your choice

2. Task Description

You are required to build an image classification pipeline using MobileNet and evaluate its performance across multiple domains.

2.1 Dataset

You may choose one of the following:

- CIFAR-10
- CIFAR-100
- Caltech101
- Intel Image Classification Dataset
- OR instructor-approved dataset

Requirements:

- Minimum 5 classes
- Balanced dataset preferred
- Proper train/validation/test split

Recommended split:

- Training: 70%
- Validation: 15%
- Test: 15%

3.2 Model Requirement

You must use MobileNet (V2 or V3) using PyTorch or TensorFlow/Keras.

Assignment #1

(Due on 14/4 at 11:59 PM)

Allowed:

- Transfer learning (recommended).
- Fine-tuning last layers.
- Input adaptation layers.

Not allowed:

- Changing backbone architecture
 - Using different CNN architectures
-

3. Input Image Domains

You must train four separate models using identical training settings.

Domain 1 — Raw Image (Spatial Domain)

- Standard RGB images
- Normal preprocessing only:
 - Resize
 - Normalize

Domain 2 — Fourier Domain

Apply:

- 2D Fast Fourier Transform (FFT)

Suggested representations:

- Magnitude spectrum
- Log magnitude

You must explain:

- What frequency domain represents
- Why it may help classification

Domain 3 — Wavelet Domain

Apply:

- Discrete Wavelet Transform (DWT)

Examples:

- Haar wavelet
- Daubechies wavelets

Assignment #1

(Due on 14/4 at 11:59 PM)

Possible inputs:

- Approximation coefficients
- Concatenated subbands

Domain 4 — Student-Chosen Domain

Students must propose and justify one domain, such as:

- Edge domain (Sobel/Canny)
- Laplacian domain
- HSV color space
- Gabor filters
- Histogram equalized images
- Learned preprocessing

Must include justification for your domain choice.

4. Experimental Constraints

All the four different domain experiments must keep identical:

- Train/test split
- Batch size
- Optimizer
- Learning rate
- Number of epochs
- Augmentation strategy (if used)

Only the input domain may change.

5. Evaluation Metrics

For each domain report:

- Training Accuracy
 - Validation Accuracy
 - Test Accuracy
 - Loss curves
 - Confusion Matrix
 - Per-class accuracy (optional bonus)
-

Assignment #1

(Due on 14/4 at 11:59 PM)

6. Required Report

Submit a technical report (PDF).

Report Structure:

Cover page with names and IDs.

1. Introduction (1–2 pages)

- Problem description
- Objective of multi-domain comparison
- Brief explanation of MobileNet

2. Image Domains

- Concept explanation
- Mathematical intuition
- Transformation pipeline
- Example transformed images

3. Experimental Setup

- Dataset description
- Preprocessing
- Model configuration
- Training parameters
- Hardware used

4. Results

Include accuracy table, confusion matrices, training curves, and observations.

5. Analysis & Discussion

- Which domain performed best and why
- Class improvements or degradation
- Frequency vs spatial insights
- Overfitting differences

Assignment #1

(Due on 14/4 at 11:59 PM)

6. Conclusion

- Key findings
 - Practical implications
 - Future work ideas
-

Submission Guidelines:

1. Teams:

- This assignment is should be done in groups of **3 to 4** students. All students must be from the tutorial groups of the **same TA**.

2. Assignment Submission:

You should submit the assignment through submission form mentioned below taking into consideration the following notes:

- You are asked to deliver a notebook (.ipynb) showing your implementation of all the required functions representing the outputs for each.
- The .ipynb/.py file containing your assignment's implementation (the notebook should be submitted showing the cells being run before and representing the output).
- Report (PDF).
- Source code.
- Trained model weights.
- README with execution steps.
- The requirements should be uploaded on your drive and provide us with the link and make sure to be accessible.
- Submit the assignment through this form:
<https://forms.gle/ghRoTCsokhMHmZf47>

Note that:

- Copying code from other teams or ChatGPT is totally prohibited. A cheating detector will be used to confirm that. Any cheating case detected will be a ZERO.
- You are not allowed to use any predefined functions for any of the requirements.
- The deadline to submit the assignment is on **Tuesday, 14th of April, 2026 at 11:59 PM**.